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SUMMARY

AI engineer embedded with users and domain experts, turning ambiguous workflow problems into adopted products. At the Department for Education, I lead discovery through rollout across regulated data and AI systems, removing manual friction and translating technical trade-offs for diverse stakeholders.

EXPERIENCE

Department for Education, HEO Data Scientist (GORS), Jan 2024 to Present

- **Designed and delivered** a Benefits Realisation Agent for PMO colleagues, translating workshop language into schema-constrained extraction, bounded generation, tool calls, validation, and retries. Reduced a routine that previously took about a week to around 90 seconds end to end; it now supports every project overseen by the division and produces documents with 95%+ accuracy.
- **Architected and led** a cross-directorate analytics platform, replacing three legacy tools and supporting roughly 300 operations and policy colleagues. Reduced time to report production from a two-week average to under five minutes for a defined output set, with selected outputs knitted into a slide deck in under one minute.
- **Delivered business continuity under a compressed deadline** by migrating a budget-forecasting model to Databricks in 11 working days against a one-month window. The in-house migration provided capability for a previously unserved departmental function without relying on a Databricks FDE or external consultants.
- **Architected** a FastAPI GraphRAG knowledge base with extended functionality that makes it queryable by other agents and tools such as Claude Code, Codex, and GitHub Copilot CLI. Community detection makes semantic and lexical relationships discoverable across a corpus, while community, sub-community, and theme retrieval reduces token usage and associated costs by 76% to 99%.
- **Rebuilt a confidential production risk-assessment system** in Python and Databricks, adding MLflow, feature-store inputs, calibrated XGBoost, drift monitoring, and Azure DevOps CI/CD; it now supports 60 colleagues conducting individual investigations in their daily operations.

Aramayo Lab, Texas A&M University, AI Consultant, Jan 2023 to Present

- **Built and continue to develop** a Scientific Workbench for genomics and bioinformatics workflows, connecting research data, literature, tools, and compute. Three labs are currently engaged, with expansion toward 12 by year-end; task evaluation is ongoing.
- **Integrated** local and hosted model serving with role-based routing and university HPC dispatch, matching execution environments to task and data sensitivity; developed modular AI-supported teaching and evaluation workflows for BIOL 213 Molecular Cell Biology using session traces.

OPEN-SOURCE AND INDEPENDENT ENGINEERING

Author and maintainer of Nuthatch, a knowledge-base GraphRAG system; Testudo, a secure containerised runtime; and Local Model Workforce, a fine-tuned specialist LLM roster for local deployment.

I also develop and maintain pi-agentic-driver, a public Pi extension set that gives multi-agent systems the controls to spawn and manage workers, steer agents away from scope creep and overengineering with code-phage, run them in monitored sandboxes with a dangerous-command kill switch, and preserve session continuity.

I integrate them into the Scientific Workbench for orchestration, retrieval, sandboxed tool use, and controlled fine-tuning and evaluation.

SKILLS

- **AI systems:** OpenAI Codex, Claude Code, Pi agent harness, MCP server authoring, context and memory management extensions, LLM orchestration, LangChain, LangGraph, RAG/GraphRAG, agentic workflows, schema-constrained extraction, tool use, LoRA fine-tuning, and local inference.
- **Production engineering:** Python, TypeScript, FastAPI, scikit-learn, PyTorch, XGBoost, Databricks, MLflow, PostgreSQL, SQL, Docker, Azure DevOps CI/CD, Plotly Dash, Electron, Node.js, React, Vite, and model monitoring.
- **FDE practice:** user discovery, requirements, technical scoping, stakeholder communication, adoption, workshops, and mentoring.

EDUCATION

- Texas A&M University, PhD Candidate, Computational Genomics (Spring 2027).
- University College London, MSci Cell and Developmental Biology (2:1), 2016.